1 The Basics

We look at G finite, k algebraically complete (just $\mathbb{C}$). We always fix a n-dimensional vector space V over k.

Definition 1. An **n-dimensional representation of** G (over k) is a homomorphism $\varphi : G \to GL_n(k)$.

Note that to define a representation I can also

- define how G acts on $v \in V$ such that the action is linear. That is, we can define $g \bullet v = w$ for any $g \in G$, and $\rho(g)$ is the linear transformation that sends v to w (hence a matrix), or
- See V as a left $\mathbb{C}G$-module (see later, mostly irrelavant to this unit, but a lot of the term is from module theory). Note that left multiplication by scalar is defined by $\bullet$.

Definition 2. Recall a group action is **faithful** if $g \bullet x = x \Rightarrow g = e$, **transitive** when the action only has one orbit.

Hence we will sometimes (confusingly) say "V is a representation" meaning that the action $\bullet$ is known. So now V above is both a linear (sub)space and a representation.

Definition 3. Two representations $\rho, \varphi : G \to GL_n(F)$ are **isomorphic** if $\varphi(g), \rho(g)$ are similiar matrices for all $g \in G$. In this situation, we write $\rho \cong \varphi$.

The key is, picking a basis we can write $\rho(g)$ as $\mathrm{diag}(\lambda_1, \lambda_2 \cdots \lambda_n)$ of eigenvalues of roots of unity.

Definition 4. Let $\rho : G \to GL(V)$ and $\varphi : G \to GL(W)$ be representations of G. Then the **direct sum** of ρ and φ is the representation $\rho \oplus \varphi : G \to GL(V \oplus W)$ corresponding to the action $g \cdot (v, w) = (g \cdot v, g \cdot w)$ of G on $V \oplus W$.

In terms of matrices, picking a basis we have $(\rho \oplus \varphi)(g) = \left(\begin{array}{c|c} A & 0 \\ \hline 0 & B \end{array}\right)$.

Definition 5. A **subrepresentation** of V is a subspace W of V such that $g \cdot w \in W$ for all $g \in G, w \in W$. A representation is **irreducible** if there are no subrepresentations other than the trivial ones ($W = \{0\}$ or V).

Clearly if $V = W \oplus W'$ then W, W' will be two subrepresentations of V.

Definition 6. If V is *not* a direct sum of two or more subrepresentations, it is called **indecomposable**.

Fortunately irreducible and indecomposable are the same in our setting of G finite and k algebraically complete.

Definition 7. Given a representation, an inner product is **G-invariant** if $\langle g \cdot v, g \cdot w \rangle = \langle v, w \rangle$ for all $v, w \in V, g \in G$.

Theorem 1. *Every representation of G has a G-invariant inner product.*

Proof. Let $\{,\}$ be any inner product on V and define $\langle v, w \rangle = \sum_{g \in G} \{g \cdot v, g \cdot w\}$, then check $\langle , \rangle$ is indeed G-invariant. Note that the finite sum in the definition uses the fact that G is finite. □

Theorem 2 (Maschke)**.** *If $W \subseteq V$ is a subrepresentation, then there exists a subrepresentation $W' \subseteq V$ such that $V = W \oplus W'$. (indecomposable $\Rightarrow$ irreducible by contrapositive)*

Proof. Let G-invariant inner product on V and let $W' = W^{\perp}$. We just need to check that W' is a subrepresentation. □

We can write (some representation) into an **irreducible representation** $V = V_1^{a_1} \oplus ... \oplus V_n^{a_n}$, where $V_1, \ldots, V_k$ are pairwise non-isomorphic irreducible representations of G.

Definition 8. $\varphi : V \to W$ is a **G-homomorphism** (G-homs) if it is linear and $g \cdot \varphi(v) = \varphi(g \cdot v)$ for all $g \in G, v \in V$. We write $\mathrm{Hom}_G(V, W)$ for the set of G-homs.

Or, equivalently, we can write $\phi \circ \rho = \sigma \circ \phi$ if ρ, σ is a the representation on V, W respectively.

Theorem 3 (Schur's Lemma)**.** *If V, W are irreducible that are not isomorphic, then* $\mathrm{Hom}_G(V, W) = \{0\}$. *Ohterwise,* $\mathrm{Hom}_G(V, W) = \{\lambda \cdot id : \lambda \in \mathbb{C}\} \cong \mathbb{C}$.

Proof. The key is: for any φ a G-hom, the kernel is always a G-hom. So the first statement follows by irreducibility. Now assume $V = W$ (i.e. φ is a isomorphism). Now pick eigenvalue λ such that $\varphi - \lambda$ id has a non-zero kernel. Therefore, the irreducibility of V implies that $\varphi - \lambda \cdot \mathrm{id} = 0$ and thus $\varphi = \lambda \cdot \mathrm{id}$. □

Note that the existence of eigenvalue relies on k be algebraically complete.

2 Theory With Characters

First corollary from Schur's Lemma is this:

Corollary. *Write* $V \cong V_1^{a_1} \oplus V_2^{a_2} \oplus \cdots \oplus V_k^{a_k}, W \cong V_1^{b_1} \oplus V_2^{b_2} \oplus \cdots \oplus V_k^{b_k}$ *as irreducible decomposition of representation. Then* $\dim \mathrm{Hom}_G(V, W) = \sum_{i=1}^k a_i b_i$.

Given $\rho : G \to GL_n(K)$, we are interested in the **character** $\chi(g) = \mathrm{Tr}\rho(g)$, note that this is invariant under conjugacy classes (since $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$). Our goal is to find the character table:

Definition 9. Let $\{\rho_1, \ldots, \rho_k\}$ be a complete set of irreducible representations of G and let χ_i be the character of ρ_i. In addition, let $C = \left\{g_1^G, \ldots, g_\ell^G\right\}$ be the set of conjugacy classes of G. The **character table** of G is the $k \times \ell$ matrix with ij-entry $\chi_i(g_j)$. We usually order the rows and columns so that $\rho_1 = \mathbb{1}$ and $g_1 = e$.

It turns out that $k = \ell$, and the character table is an "orthonormal" matrix (with respect to a different inner product in rows and columns).

Definition 10. Let G be a finite group and let $\mathscr{C}$ be the set of conjugacy classes of G. Let $\rho : G \to \mathrm{GL}(V)$ and $\sigma : G \to \mathrm{GL}(W)$ be representations of G, with corresponding characters χ and ψ. We define

$\langle \chi, \psi \rangle := \frac{1}{|G|} \sum_{g \in G} \chi(g)\overline{\psi(g)}$.

Theorem 4 (Row Orthogonality)**.** *If* $V \cong V_1^{a_1} \oplus \cdots \oplus V_k^{a_k}$ *and* $W \cong V_1^{b_1} \oplus \cdots \oplus V_k^{b_k}$*, then* $\langle \chi_V, \chi_W \rangle = \sum_{i=1}^k a_i b_i$.

This is very useful: if one of V, W (say V) is irreducible, this inner product tells me how many Vs are in W (in the irreducible decomposition), Also, doing $\langle \chi_V, \chi_V \rangle$ also give a hint of the multiplicities (as a sum of square).

Theorem 5 (Column Orthogonality)**.** $x, y \in G$ *then* $\sum_{\chi \in \mathrm{Irr}(G)} \chi(x)\overline{\chi(y)} = \begin{cases} |C_G(x)| & \text{if } x \text{ and } y \text{ are conjugate} \\ 0 & \text{otherwise} \end{cases}$,

remembering that

$$|C_G(x)| = \frac{|G|}{|\text{The conjugacy class containing } x|}$$

by orbit-stablizer.

Now we present some common construction and their representations.

- The direct sum - corresponds to the "pointwise" sum of characters.
- The group algebra $\mathbb{C}G$ (with basis as the element of g), it is a vector space, the action is defined by left multiplication: $g \bullet \lambda h := \lambda gh$ (and extending linearly). It has a representation of $\mathbb{C}G \cong \bigoplus_{i=1}^k V_i^{\dim V_i}$. In particular, $|G| = \sum_{i=1}^k (\dim V_i)^2$. This depends on $\mathrm{Hom}_G(\mathbb{C}G, V) \cong V$ and there is only one map you can write down. This actually shows there can only be finitely many irreducible representations in a finite group G, and the last one can always be found by subtracting scalar multiple and dividing the dimension of the known representations. The character for $\mathbb{C}G$ is $(|G|, 0, 0, 0, \cdots 0)$.
- The dual V corresponds to complex conjugating a character. Note that the conjugacy classes are self-inverse if and only if conjugating a character gives you the same character (i.e. the character is real-valued).
- Multiplying two character "point-wise" corresponds to the tensor product $V \otimes W = \mathrm{Hom}(V^*, W)$.
- The tensor product $V \otimes V$ can be broken down into direct sum $S^2V \oplus \Lambda^2 V$, this corresponds to representations:

$$\chi_{S^2V}(g) = \frac{\chi_V(g) + \chi_V(g^2)}{2}, \chi_{\Lambda^2 V}(g) = \frac{\chi_V(g) - \chi_V(g^2)}{2}.$$

Definition 11. Given a normal subgroup $N \lhd G$ and a representation ρ on G/N, one can **lift** ρ to a representation on G by defining $\widehat{\rho} := \rho \circ \pi$, where $\pi : G \to G/N$ is the canonical projection.

If ρ is irreducible, then $\widehat{\rho}$ is irreducible as well?

3 Computing the Character Table

One might be able to get the most of the character table by computing a lot of inner products using the techniques above and doing the "$\mathbb{C}G$" trick, but there are more tricks which are available to us. First, there is a routine way to find 1-dim irreducibles.

Definition 12. The **derived/commutator subgroup** of G, denoted as G' is the subgroup generated by all the commutators $[x, y] = xyx^{-1}y^{-1}$.

Theorem 6. *For any normal N, G/N is abelian $\Leftrightarrow$ $G' \leq N$, that is, G' is the minimal one to have this property.*

Theorem 7 (1-dimesnional representation Classification)**.**

- *Every irreducible representation of a finite abelian group G is 1-dimensional, and there are $|G|$ such representations in total.*
- *Every finite group G has precisely $|G : G'| = |G|/|G'|$ 1-dimensional representations; they are all lifted from G/G'.*

G is finite so it can be thought of as a product of C_{p_i}s (p_i prime). The 1-dimensional representations are just generators taking p_i-roots of unity.

Another way to construct representation is to think of a group action. Let G acts on a finite set X (on the left).

Theorem 8. *There is a character $\chi(g) = Fix(g) = \#\{x \in X : gx = x\}$.*

This is a character on $\mathbb{C}X$ (just like the group algebra $\mathbb{C}G$). In particular, it is positive integer-valued, so it always gives something larger than 1 if we consider the inner product with $\mathbb{1}$ (In fact, it is the number of orbits on X by G). This is another good place to start thinking about getting higher-dimension irreducibles.

Another way that works with any subgroup $H \leq G$ is induction/restriction. The catch is they are left/right adjoint in some ways. Let start with the easy one.

Definition 13. Given a representation ρ on G, then for any $H \leq G$ the **restricted representation** is $\mathrm{Res}^G_H \rho(h) := \rho|_H$ on H.

Nothing really special on the character side, but note that going into subgroups the conjugacy classes might split (e.g. A_5, the 5-cycles splits into 2).

But going from H to G is quite hard:

Definition 14. Let $\sigma : H \to GL(U)$ be a representation on H, then the **induced representation** $\mathrm{Ind}^G_H \sigma$ can be define to have the vector space $V = \mathrm{Hom}_H(\mathbb{C}G, U)$ and a G-action on $V : (g \bullet \phi)(x) := \phi(gx)$

From here it is hard to see anything useful, but there is a formula for the character of the induction.

Theorem 9. *Let ψ be a character of H, write $G = Hx_1 \cup Hx_2 \cdots \cup Hx_n$ as a disjoint union of right cosets (also called **right transversal**). Then define*

$$\psi^0(g) = \begin{cases} \psi(g) & \text{if } g \in H \\ 0 & \text{if } g \notin H \end{cases}, \text{ then we have } \mathrm{Ind}^G_H \psi(g) = \sum_{i=1}^{n} \psi^0\left(x_i g x_i^{-1}\right)$$

In particular, if $H \lhd G$, then

$$\mathrm{Ind}^G_H \psi(g) = \begin{cases} 0 & \text{if } g \notin H \\ \sum_{i=1}^{n} \psi\left(x_i g x_i^{-1}\right) & \text{if } g \in H \end{cases}$$

Note that it follows induction multiplies a character's dimension by the index of the group.

Theorem 10 (Frobenius Reciprocity)**.** $\left\langle \mathrm{Ind}^G_H \psi, \chi \right\rangle_G = \left\langle \psi, \mathrm{Res}^G_H \chi \right\rangle_H$.

Below are some random facts to check whether your character table makes sense:

- $\chi(e)$ is the dimension of the representation associated to χ, and it always divide $|G|$.
- Every entry of a character is the sum of roots of unities, hence an algebraic integer.